Master: Amylase Rate, Graph and Topic 1 Consolidation

Today I can… □ Show your working: relative rate = 1 ÷ time. Use the unit s^{-1}. □ Read or plot how pH links to rate. Say only what the data supports. □ Name a limit of the method. Link it to a way to improve it. Then show how digestion differs from diffusion.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

TA brief

Access and independence

At Independent, one adult supports access to Supported while the other holds the group; swap across the week. Wait, gesture, cue, model one step, then try a fresh step independently. Give VF, then one precise NS+; wait for a real pupil response R and // edit. Use ? for read-back. WS and I describe actual support; E requires target-matched evidence.

During We do, the lesson teacher is the named audience. Offer an optional request for reading help, fewer pieces, clearer labels, adult-directed handling or thinking time. Try the change and check its effect. Passing and private requests are valid.

Before the lesson, scan the books for five minutes. Notice previous feedback, pupil responses and where the learning stopped.

Ask the cover teacher and TAs what was actually taught, who was present and what changed for individual pupils.

Check the roll against the books. New arrivals have no prior evidence; start from what every pupil shows now.

Existing NS+ is history. Do not chase, back-mark or retrospectively close it. Begin fresh feedback from this lesson.

After the check, one read, once, shapes Autumn 2. Read each success criterion with its access support, choose the next route, then teach; do not reopen earlier feedback.

Show relative-rate working using 1 ÷ time with the unit s^{-1}. Green: accurate explanation with evidence independently → Aspire application. Amber: part-correct or secure after one content cue → standard with one worked example. Red: no secure evidence after an access adjustment → supported re-model and a fresh check. Record access support separately; do not average criteria.

Read or plot the pH–rate pattern and limit the conclusion to the evidence. Green: accurate explanation with evidence independently → Aspire application. Amber: part-correct or secure after one content cue → standard with one worked example. Red: no secure evidence after an access adjustment → supported re-model and a fresh check. Record access support separately; do not average criteria.

Link a method limitation to an improvement, then distinguish digestion from diffusion. Green: accurate explanation with evidence independently → Aspire application. Amber: part-correct or secure after one content cue → standard with one worked example. Red: no secure evidence after an access adjustment → supported re-model and a fresh check. Record access support separately; do not average criteria.

The criterion profile selects the next routes: begin with the lowest insecure dependency, teach one example, then check fresh evidence and fade the support. A non-response prompts an access check before a science judgement.

Master: Amylase Rate, Graph and Topic 1 Consolidation

Today I can… ☐ Show your working: relative rate = 1 ÷ time. Use the unit s^{-1}. ☐ Read or plot how pH links to rate. Say only what the data supports. ☐ Name a limit of the method. Link it to a way to improve it. Then show how digestion differs from diffusion.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

TA brief · continued

Preparation and practical safety

Use the supplied models, cards, data and booklet. This lesson plans only. During the later practical, handle reagents only with the centre-approved method, prepared equipment, local risk assessment and trained supervision. The supplied-observation route needs no reagents and teaches the same interpretation; it does not certify physical practical completion. Do not infer concentrations, heating arrangements or disposal from lesson cards.

Destination link (Launch): rate calculations from timed observations are [QUAL] Biology exam skills and the everyday work of laboratory technicians, quality-control testers and biomedical science routes at Level 3 and beyond.

Choose a route at each task using what the pupil shows. Switching adds no task, record or label.

Risk-note controls cross-checked against publicly accessible [SYS] [QUAL] Enzymes and [QUAL] Iodine on 26 September 2026. Teacher/technician must authorise the local method, concentrations, quantities, PPE and disposal using current member guidance. The full member-only [QUAL] method and [SYS] were not accessible for this check.

Use minimum enzyme and iodine concentrations and quantities. Prevent enzyme powder aerosol; do not spray solutions. Follow eye-protection and solvent-specific iodine controls.

Illustrative or supplied data is reasoning practice only and is never logged as practical completion. Log only an observed, authorised physical contribution; the teacher decides whether it meets the relevant course requirement.

Prompt ladder

Prompt ladder for adults

1. Point to the relevant place / instruction.
2. Ask one short question: "What is it asking you to find?"
3. Rephrase the instruction in simpler language.
4. Provide a scaffold or limited choice.
5. Model the first step only, then return the task to the pupil.

Master: Amylase Rate, Graph and Topic 1 Consolidation

Today I can… □ Show your working: relative rate = 1 ÷ time. Use the unit s^{-1}. □ Read or plot how pH links to rate. Say only what the data supports. □ Name a limit of the method. Link it to a way to improve it. Then show how digestion differs from diffusion.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

TA brief · continued

Planned adaptations

Cognition and learning: Arrival: reveal one clue at a time; model a reasoning step, then fade the cue for a fresh example.

Communication and interaction: We do: allow thinking time, paired rehearsal, pointing, sign or AAC before giving a reason.

SEMH: Independent: agree a quiet pause and one return step; correct privately; accept a pass on a personal example.

Sensory and physical: Models: offer reduced motion, paused diagrams, enlarged prepared pieces, seated work or adult-directed placement.

EAL (language access, not SEN): Arrival: pre-teach topic words beside the diagram; permit any-language planning and a sentence stem for the same science.

Capture E against the pupil's own [SYS] target on the school's evidence app; the [SYS] carries their EHCP-linked and [SYS] targets, so one record serves every plan.

Follow the pupil's [SYS] where it differs from these routines.

00–03 · Arrival task

At arrival, say the spoken line on the marking card once. Use genuine last lesson evidence for the spoken inheritance. This arrival is the only calibration; no separate pre-check or recovery task. IF NEW — 0.5 min read clues; 1.5 min respond; 1 min sample and reveal. IF REVISION — 0.5 min orient; 1.5 min respond; 1 min sample and reveal.

03–06 · Starter · reverse the time

At about minute eight, choose one same-lesson adjustment from the arrival: hold calculator modelling longer, use supported graph placement, switch the current task route or add the repeat-evidence stretch card. IF NEW — 0.5 min compare times; 1 min predict rates; 1.5 min explain direction. IF REVISION — 0.5 min predict; 1.5 min justify using reciprocals; 1 min check conditions.

06–10 · I do · time becomes relative rate

Use 1/t only. Show 1/120 = 0.00833 s^{-1}; discuss sufficient rounding without pretending interval-based observations are exact instantaneous rates. IF NEW — 2.5 min model formula, substitution and unit; 1.5 min connect the result to its graph point. IF REVISION — 0.5 min orient; 2.5 min pupils predict and check calculations; 1 min compare direction.

Master: Amylase Rate, Graph and Topic 1 Consolidation

Today I can… □ Show your working: relative rate = 1 ÷ time. Use the unit s^{-1}. □ Read or plot how pH links to rate. Say only what the data supports. □ Name a limit of the method. Link it to a way to improve it. Then show how digestion differs from diffusion.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

TA brief · continued

10–14 · We do · build an evidence chain

Named involvement opportunity. Card-then-destination works by keyboard and touch. Ask whether each statement comes from data, biological knowledge or a judgement of the method. IF NEW — 1 min read categories; 2 min classify and justify; 1 min respond to a pupil's reasoning. IF REVISION — 0.5 min orient; 2.5 min classify with evidence; 1 min compare the claim boundary.

14–17 · I do · read the graph honestly

Label pH 7 as highest in the first series, among values tested. Do not silently average the repeat, impose a universal pH 7 optimum or treat a fall in activity as proof of irreversible denaturation. IF NEW — 1.5 min reveal the five-point pattern; 0.5 min add the separate repeat; 1 min distinguish data from explanation. IF REVISION — 0.5 min inspect; 1.5 min qualify the peak with the repeat; 1 min compare scope and uncertainty.

17–21 · We do · repair and transfer

Model one concise repair, then name the two processes. This supplies the Topic 1 transfer before independent work, so a missed earlier topic is not a barrier. IF NEW — 1 min inspect the claim and process definitions; 2 min repair and label; 1 min check. IF REVISION — 0.5 min inspect; 2.5 min repair with a reason; 1 min check the transfer.

21–34 · You do

Use pre-scaled axes with three points already shown and two points to add; do not spend the block redrawing axes or copying the full table. Five short responses retain calculation, graph, explanation, evaluation and transfer within fifteen minutes. Supported pupils may direct plotting and dictate the same reasoning. IF NEW — 2 min orient the calculations and graph; 11 min complete the five responses; 2 min use one precise feedback point. IF REVISION — 1 min orient; 12 min complete the five responses; 2 min use one precise feedback point.

34–37 · Review and improve

Give one specific VF and one NS+; wait for a real // edit and hear the pupil explain the change.

37–40 · Exit ticket

Use one book response OR one ticket. Read once after last lesson to shape the plan for Autumn 2, since half term follows. No retrospective closure or additional pupil recording. IF NEW — 1 min read evidence; 2 min respond; 1 min pupil finishes the same response. IF REVISION — 0.5 min orient; 2.5 min respond; 1 min pupil finishes the same response.

Master: Amylase Rate, Graph and Topic 1 Consolidation

Today I can… □ Show your working: relative rate = 1 ÷ time. Use the unit s^{-1}. □ Read or plot how pH links to rate. Say only what the data supports. □ Name a limit of the method. Link it to a way to improve it. Then show how digestion differs from diffusion.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

TA brief · continued

Planning and evidence

Say once: “I haven't seen what you've been doing… I start from what you show me”.

Lesson code: SCI-LAUNCH-A1-W08-L3. SoW reference: LAUNCH Science, Autumn 1 return block; L3.

Exact SoW outcome: I can calculate amylase rate from endpoint time, plot and interpret the pH pattern, evaluate the method, and apply Topic 1 knowledge to exam-style evidence.

Source placement: enzyme activity enrichment consolidates Topic 1; the workbook has no separate Autumn 1 Week 8 row. This is the supplied return-period mapping, not a newly invented workbook objective.

This lesson: I can calculate and graph relative amylase rates, judge what the evidence supports, and use enzyme and transport ideas in an exam-style explanation.

Evidence (scheme of work): [AWARD] [QUAL] Paper 1 ([QUAL]) topic tests; [AWARD] [QUAL] science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec). Evidence (this lesson): Show relative-rate working using 1 ÷ time with the unit s^{-1}.; Read or plot the pH–rate pattern and limit the conclusion to the evidence.; Link a method limitation to an improvement, then distinguish digestion from diffusion.

Evidence location: the pupil’s explanation under the book strip, with route, actual support, review edit // and exit response. Keep raw observations distinct from illustrative data.

Label evidence on the school's evidence app with this code: SCI-LAUNCH-A1-W08-L3. Record the observed science and prompt level; clicks and access aids do not earn a marking code.

Staff: confirm the planned adaptations actually used and record any changes below. [SYS] strategies are linked to actual tasks above.

60-second recap of the previous lesson: Samples were checked at fixed times. Blue-black detected starch; orange-brown did not. At pH 7, 60 s was blue-black and 90 s was orange-brown. Record the first observed endpoint, 90 s, while stating the window after 60 s and by 90 s. Keep missed samples as gaps.

Must teach core: I can calculate and graph relative amylase rates, judge what the evidence supports, and use enzyme and transport ideas in an exam-style explanation. Shorten or merge: take one shared explanation; preserve independent evidence, review and exit. Next lesson: revisit any insecure criterion before extending the topic. Dependency: Show relative-rate working using 1 ÷ time with the unit s^{-1}.

Feedback and evidence

No cold calling; nobody is asked to read difficult text aloud.

Staff mark I, S or C beside this task.